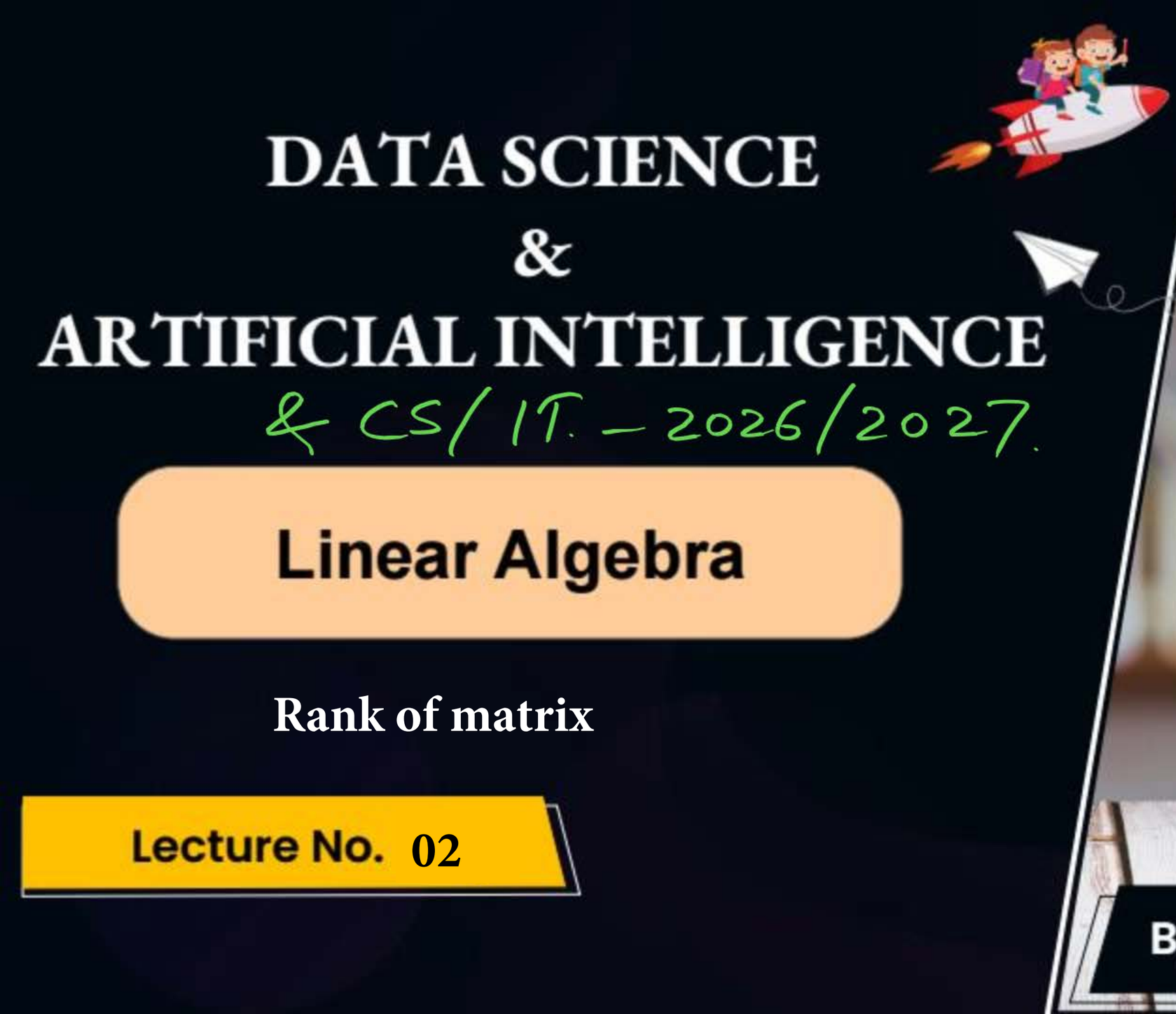
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Rank of matrix

Lecture No. 02



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

RANK & it's Properties

Topics to be Covered



Topic

1. ECHELON FORM
2. Nature of vectors

(Ex) $A=\begin{bmatrix}2 & -3 & 1\\ 4 & 2 & 0\end{bmatrix}_{2\times3}$, $B=\begin{bmatrix}3 & -2\\ 1 & 4\\ -1 & 2\end{bmatrix}_{3\times2}$ then $\begin{cases}Tr(AB)=?=2\\ Tr(BA)=?=2\end{cases}$

Sol: $Tr(A)=DNE$, $Tr(B)=DNE$

$$AB=\begin{bmatrix}2 & -3 & 1\\ 4 & 2 & 0\end{bmatrix}\begin{bmatrix}3 & -2\\ 1 & 4\\ -1 & 2\end{bmatrix}=\begin{bmatrix}2 & -14\\ 14 & 0\end{bmatrix}_{2\times2}\Rightarrow Tr(AB)=2$$

$$BA=\begin{bmatrix}3 & -2\\ 1 & 4\\ -1 & 2\end{bmatrix}\begin{bmatrix}2 & -3 & 1\\ 4 & 2 & 0\end{bmatrix}=\begin{bmatrix}-2 & -13 & 3\\ 18 & 5 & 1\\ 6 & 7 & -1\end{bmatrix}_{3\times3}\Rightarrow Tr(BA)=2$$

Conclusion: If $A_{m\times n}$ & $B_{n\times m}$ then $\boxed{Tr(AB)_{m\times m}=Tr(BA)_{n\times n}}$

Echelon Form (Triangular Form) → Mat $A_{m\times n}$ is said to be in Echelon form, if

① Number of Zeros **before** the 1st Non Zero element in a Row should be in an Increasing order in the subsequent Rows.

② All Zero Rows (if exist) should occur after Non Zero Rows.

Note: ① ρ(Echelon form) = Number of Non Zero Rows.

② **Any** Mat of $A_{m\times n}$ Can be converted into an E-form by using E-operations. But take Care

" we should apply **only** E-Row operations while converting our Mat into an E-Form " (As per our syllabus)

Flowchart to Convert given Mat into an E-Form →

(1) Make a_{11} unity (Not Compulsory but advisable)

(2) Make all the elements that lies below a_{11}, Zero using E-Row Operations

(3) Make a_{22} unity (Not Compulsory but advisable)

(4) Make all the elements that lies below a_{22}, Zero using E-Row Operations

(5) Make a_{33} unity & so on - - - - - -

Sp Note- In E-Form Take Care; $a_{21} = 0$

e.g. $\begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 2 & 1 & -3 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 4 \end{bmatrix}$, $\begin{bmatrix} 2 & -1 & 4 & 1 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 4 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

✓ $\rho(A)=4$ ✓ $\rho(A)=3$ ✓ $\rho(A)=3$ ✓ $\rho(A)=3$

e.g. $\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 5 & -1 & 2 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 3 & -5 & 4 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

✗ $\rho(A)=?$ ✗ $\rho(A)=?$

∴ Last two Matrices are not in Echelon Form so 1st we will convert them into an Echelon Form & then we will Find their Rank.

Qe Convert the following Mat into an E-Form & Hence find it's Rank.

$$A=\begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \end{bmatrix}_{6\times 6}$$

(M-I) Using Observation, Consider a submatrix

$$A_1=\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}_{3\times 3}$$

By deleting R_2, R_4, R_5 & C_4, C_5, C_6

$\because |A_1| \neq 0$ i.e A_1 is Non sing $\Rightarrow \rho(A) =$ three

(M-II) $R_2 \leftrightarrow R_3$

$$\begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_6} \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = A_1$$

which is in E-Form, $\therefore \rho(A_1) =$ No of Non Zero Rows $= 3$

$\because A \sim A_1 \Rightarrow \rho(A) = 3$

Q: $A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}_{5\times5}$ $\xrightarrow{R_4 \to R_4 + R_1}$ $\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$ $\xrightarrow{R_2 \leftrightarrow R_3}$ $\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$

$\xrightarrow{R_4 \to R_4 + R_2}$ $\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$ $\xrightarrow{R_4 \to R_4 + R_3}$ $\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$

$\xrightarrow{R_5 \to R_5 + R_4}$ $\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{5\times5}$ $\Rightarrow$ (Non-Singular U.T.M of 4×4)

$= E.Form$ $\&o\ \rho(A) = 4$

M-II

2015
2M

$$A=\begin{bmatrix}1&-1&0&0&0\\0&0&1&-1&0\\0&1&-1&0&0\\-1&0&0&0&1\\0&0&0&1&-1\end{bmatrix}\xrightarrow[R_4\leftrightarrow R_5]{R_2\leftrightarrow R_3}\begin{bmatrix}1&-1&0&0&0\\0&1&-1&0&0\\0&0&1&-1&0\\0&0&0&1&-1\\-1&0&0&0&1\end{bmatrix}$$

$$\xrightarrow{R_5\to R_5+(R_1+R_2+R_3+R_4)}\begin{bmatrix}1&-1&0&0&0\\0&1&-1&0&0\\0&0&1&-1&0\\0&0&0&1&-1\\0&0&0&0&0\end{bmatrix}_{5\times 5}$$ = E-form $\because \rho(A)=4$

M-III

$$A=\begin{bmatrix}1&-1&0&0&0\\0&0&1&-1&0\\0&1&-1&0&0\\-1&0&0&0&1\\0&0&0&1&-1\end{bmatrix}\xrightarrow{R_2\leftrightarrow R_4}\begin{bmatrix}1&-1&0&0&0\\-1&0&0&0&1\\0&1&-1&0&0\\0&0&1&-1&0\\0&0&0&1&-1\end{bmatrix}$$ (Not in E-Form)

$$\xrightarrow{R_2\to R_2+(R_1+R_3+R_4+R_5)}\begin{bmatrix}1&-1&0&0&0\\0&0&0&0&0\\0&1&-1&0&0\\0&0&1&-1&0\\0&0&0&1&-1\end{bmatrix}\cdots\cdots\begin{bmatrix}1&-1&0&0&0\\0&1&-1&0&0\\0&0&1&-1&0\\0&0&0&1&-1\\0&0&0&0&0\end{bmatrix}$$ = E-Form.

Q8 $A = \begin{bmatrix} 3 & -2 & 1 & 4 \\ 2 & 1 & -3 & 0 \\ -1 & 2 & 4 & 1 \\ -4 & -1 & -2 & -5 \end{bmatrix}_{4\times4}$ $\xrightarrow{R_1 \leftrightarrow R_3}$ $\begin{bmatrix} -1 & 2 & 4 & 1 \\ 2 & 1 & -3 & 0 \\ 3 & -2 & 1 & 4 \\ -4 & -1 & -2 & -5 \end{bmatrix}$

$\xrightarrow{\substack{R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 + 3R_1 \\ R_4 \to R_4 - 4R_1}}$ $\begin{bmatrix} -1 & 2 & 4 & 1 \\ 0 & 5 & 5 & 2 \\ 0 & 4 & 13 & 7 \\ 0 & -9 & -18 & -9 \end{bmatrix}$ $\xrightarrow{\substack{R_2 \to R_2 - R_3 \\ R_4 \to -\frac{1}{9}R_4}}$ $\begin{bmatrix} -1 & 2 & 4 & 1 \\ 0 & 1 & -8 & -5 \\ 0 & 4 & 13 & 7 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

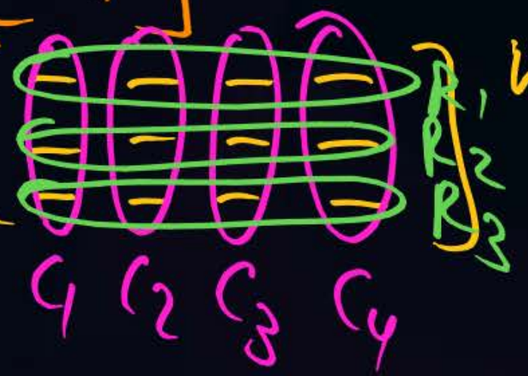
$\xrightarrow{\substack{R_3 \to R_3 - 4R_2 \\ R_4 \to R_4 - R_2}}$ $\begin{bmatrix} -1 & 2 & 4 & 1 \\ 0 & 1 & -8 & -5 \\ 0 & 0 & 45 & 27 \\ 0 & 0 & 10 & 6 \end{bmatrix}$ $\xrightarrow{R_3 \to \frac{1}{45}R_3}$ $\begin{bmatrix} -1 & 2 & 4 & 1 \\ 0 & 1 & -8 & -5 \\ 0 & 0 & 1 & 27/45 \\ 0 & 0 & 10 & 6 \end{bmatrix}$ $\xrightarrow{R_4 \to R_4 - 10R_3}$ $\begin{bmatrix} -1 & 2 & 4 & 1 \\ 0 & 1 & -8 & -5 \\ 0 & 0 & 1 & 27/45 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$A \sim \begin{bmatrix} -1 & 2 & 4 & 1 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & 1 & 3/5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\Rightarrow \rho(A) =$ No. of Non Zero Rows $=$ three.

NATURE of VECTORS

ordered n-tuple (n-Dim. Vector) :→ Any ordered set of n numbers is called n-tuple.
Generally it is represented in the form Column Mat (But we can also represent it in the form of Row Mat). ie

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \cong \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_n \end{bmatrix} \cong n\text{-dim vector.}$$



has 3 Row vectors & 4 Column vectors

eg ordered pair $= (x_1, x_2) = x_1\hat{i} + x_2\hat{j} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$ 2-vector

(Point in 2D)

eg ordered Triplet $= (x_1\, x_2\, x_3) = x_1\hat{i} + x_2\hat{j} + x_3\hat{k} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$ 3-vector

(Point in 3D).

eg ordered quadruple $= (x_1\, x_2\, x_3\, x_4) = ? = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} =$ 4-vector

$\vdots$ $\qquad$ $\vdots$

ordered n-tuple $= (x_1 x_2 \cdots x_n) = ? = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_4 \end{bmatrix} =$ n-vector

Dot Product: $X=\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$, $Y=\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$ then $\boxed{X \cdot Y = X^T Y} = a_1b_1 + a_2b_2 + \dots + a_nb_n$

i.e. $X \cdot Y = X^T Y = \underset{1\times n}{[a_1\ a_2\ a_3 \dots a_n]} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}_{n\times 1} = \underset{1\times 1}{[(a_1b_1 + a_2b_2 + \dots + a_nb_n)]}$

Also Calculate $X \cdot Y = ? = \underset{n\times 1}{[\ ]}\ \underset{n\times 1}{[\ ]}$ = Not Defined

also Calculate $X^2 = ? = X \cdot X = \underset{n\times 1}{[\ ]}\ \underset{n\times 1}{[\ ]}$ = N.D.

Norm of Vector (Length): $\boxed{\|X\| = \sqrt{X^T X}} = \sqrt{a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2}$

Normalised Vector (unit vector) ⟶ if $(\|x\| = 1)$ $\Leftrightarrow$ then (x is called Normalised vector)

orthogonal vectors ⟶ if $(x \cdot y = 0)$ $\Leftrightarrow$ then (x & y are called orthogonal vector)

Note: three vectors x, y, z are orthogonal if they are pairwise orthogonal. i.e $x \cdot y = y \cdot z = z \cdot x = 0$.

orthonormal vectors ⟶

if $(x \cdot y = 0$ & $\|x\| = 1, \|y\| = 1)$ $\Leftrightarrow$ then (x & y are called orthonormal vectors)

i.e x & y are called orthonormal vectors if they are orthogonal as well as they must Normalised also.

(*) **Column vectors of an orthogonal Mat are orthonormal vectors.**

(Learn)

for eg Consider an orthogonal Mat

$$A = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

C_1 C_2 C_3

then

$$C_1 \cdot C_2 = C_2 \cdot C_3 = C_3 \cdot C_1 = 0$$

$$\|C_1\| = \|C_2\| = \|C_3\| = 1$$

Hence justified.

Q. Check the orthogonality of vectors $A=\{(1\ 2\ 1)^T, (2\ 1\ -4)^T, (3\ -2\ 1)^T\}$ & Also check their orthonormality ?

Sol: $A=\{X_1\ X_2\ X_3\}$ where $X_1=\begin{bmatrix}1\\2\\1\end{bmatrix}$, $X_2=\begin{bmatrix}2\\1\\-4\end{bmatrix}$, $X_3=\begin{bmatrix}3\\-2\\1\end{bmatrix}$

Now, $X_1 \cdot X_2 = (1)(2)+(2)(1)+(1)(-4) = 0$

$X_2 \cdot X_3 = (2)(3)+(1)(-2)+(-4)(1) = 0$

$X_3 \cdot X_1 = (3)(1)+(-2)(2)+(1)(1) = 0$

ie X_1, X_2, X_3 are orthogonal vectors.

(ii) $\|X_1\| = \sqrt{1^2+2^2+1^2} = \sqrt{6}$ ie X_1, X_2, X_3 are not orthonormal vectors

$\|X_2\| = \sqrt{2^2+1^2+(-4)^2} = \sqrt{21}$

$\|X_3\| = \sqrt{3^2+(-2)^2+1^2} = \sqrt{14}$

Q: If $A = \begin{bmatrix} \frac{1}{9} & -\frac{4}{9} & \frac{8}{9} \\ \frac{8}{9} & \frac{4}{9} & \frac{1}{9} \\ \frac{\alpha}{9} & -\frac{7}{9} & \frac{\beta}{9} \end{bmatrix}$ is an Orthogonal Mat then $\alpha+\beta = ?$

(a) 4 (b) −4 (c) 0 (d) 16

C_1 C_2 C_3

Acc to Property of O-Mat; we have $C_1 \cdot C_2 = 0$ & $C_2 \cdot C_3 = 0$

$\frac{1}{81}[-4+32-7\alpha] = 0$ & $\frac{1}{81}[-32+4-7\beta] = 0$

$28 - 7\alpha = 0$ & $-28 - 7\beta = 0$

$\alpha = 4$ & $\beta = -4$

THANK - YOU

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